

Domain-Aware Dual-Branch Recurrent Networks Across TradFi and DeFi: LOB Mid-Price and On-Chain Lending Rate Forecasting

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Abstract

Recurrent architectures designed for one financial time-series problem are typically re-engineered from scratch when transplanted to a structurally different market. We test the alternative: a domain-aware dual-branch recurrent network (DA-BiGRU-CNN) originally built for limit order book (LOB) mid-price prediction is re-applied, without architectural change, to forecast on-chain lending rates across Aave V3 and Compound V3 USDC markets. The two BiGRU branches and the multi-scale Conv1d fusion head are re-specialised by identifying the domain-natural decomposition pair: in LOB the pair is (price, volume) augmented with microstructure features; in DeFi it is (rate residual $\varepsilon = r - f_{\text{kink}}(u)$, utilization u with context). The same composite loss $\alpha \text{MSE} + \beta (1 - \rho_w) + \gamma Q_{90}$ is the trained objective in both domains. In the prior LOB study the network attains weighted-Pearson 0.266 on test, beating a 205-feature LightGBM (0.168) by 58%. In the present DeFi application, trained on 12,895 hourly bars at 98.5% panel coverage from January 2026 to April 2026, the forecaster feeds a 4-factor MCDM allocator; a paired monthly-Sharpe bootstrap returns a point-estimate Sharpe gain of +1.667 with a 95% confidence interval that includes zero ($p = 0.121$). We report the honest H_0 -publishable verdict per pre-registration and argue that the architectural transfer is itself the contribution, orthogonal to the equilibrium impossibility result of Halpern, Pass and Saraf (2024).

Keywords: ICICPE, DeFi, MCDM, BiGRU-CNN, cross-domain transfer

1. Introduction

Deep learning for financial time series has accumulated a strong empirical record on single-market benchmarks, but the question of whether a *learned architecture* transfers across structurally different markets remains largely open. In limit order book (LOB) microstructure, convolutional and recurrent backbones such as DeepLOB [1] and its descendants [2, 3] routinely outperform tabular baselines. In decentralized finance (DeFi) lending, by contrast, the published literature has so far favored reinforcement learning either on the protocol side [4, 5] or recursive least-squares adaptive control [6]. A practitioner who has invested in a domain-aware recurrent stack for one of these markets faces an honest question: must the model be re-engineered from scratch for the other, or is the architectural template itself portable?

The two markets are not obviously compatible. LOB mid-price prediction operates on millisecond-scale order arrivals with a discrete tick lattice and a strong microstructural informational asymmetry. DeFi lending rates, by contrast,

are set deterministically from on-chain state via a known piecewise-linear kink function $f_{\text{kink}}(u)$ of utilization u ; they update hourly at most and move primarily because of cointegration between protocols [7] and slow demand-driven repricing [8]. The two are noisy multivariate time series with regime-dependent dynamics and a weighted-error preference for large moves, but their feature topologies, decision horizons, and competitor adversaries differ qualitatively.

Central thesis. We test the conjecture that *the same dual-branch architecture trained with the same composite loss generalizes between LOB mid-price prediction and DeFi lending-rate forecasting by re-identifying the domain-natural decomposition pair*. In LOB the pair is (price, volume) augmented with 21 microstructure features [9]; in DeFi it is (rate residual $\varepsilon = r - f_{\text{kink}}(u)$, utilization with context). The two BiGRU branches, multi-scale Conv1d fusion head, and the composite loss $\alpha \text{MSE} + \beta (1 - \rho_w) + \gamma Q_{90}$ are unchanged between the two specialisations; only the input subspaces and their dimensionality are re-mapped to the target domain.

Contributions. The paper makes four contributions.

- **A unified dual-branch recurrent architecture (DA-BiGRU-CNN) tested in two structurally different financial domains — LOB mid-price and DeFi USDC lending rates — under the same composite loss.**
- **A new empirical study: to our knowledge the first published forecast-driven multi-criteria allocator across Aave V3 and Compound V3 USDC pools, trained on 12,895 hourly bars spanning January 2026–April 2026 at 98.5% panel coverage, with regime structure documented at full coverage.**
- **An honest negative-result protocol.** We pre-register the headline H1 ($\Delta\text{SHARPE} \geq 0.2$) and report $\Delta\text{SHARPE} = +1.667$ with the bootstrap 95% CI $[-6.459, +11.031]$ and p -value 0.121 — failing to reject H_0 under the binding $n_{\text{months}} = 4$ test window, exactly as the pre-registration permitted.
- **Two upstream pull requests to the open-source fractal-defi framework [10]:** a Compound V3 Mesari loader and a `BaseLendingAllocationStrategy` abstraction with cooldown infrastructure — both released alongside the paper for reproducibility.

Outline. Section 2. reviews LOB microstructure DL, DeFi lending mechanics, the impossibility result of [11], and related transfer-learning literature. Section 3. gives the domain-agnostic architecture and its two specialisations. Section 4. summarises the LOB results we treat as prior work. ?? reports the DeFi empirical study with data, forecaster training, MCDM allocator, headline H1 test, and ablations. Section 6. states what transfers and what does not, and Sections 7. and 8. discuss honest limitations and close.

2. Background and Related Work

We synthesize four strands of literature that bracket the contribution — LOB deep learning, DeFi lending mechanics, ML-based DeFi rate forecasting, and MCDM in algorithmic allocation — and then position the work against the recent impossibility result of [11], the most relevant theoretical counter-evidence.

2.1 LOB microstructure deep learning

Modern LOB mid-price prediction has converged on convolutional-recurrent backbones. DeepLOB [1] established

the CNN+LSTM template on the FI-2010 benchmark [12]; [13] demonstrated cross-stock universal features under deep learning; [14, 2, 15] extended the family with attention and bilinear mechanisms. The author’s prior dual-branch architecture [9] is the immediate predecessor to the present work and we recap its key results in Section 4.. Recent crypto-LOB work [16, 3] reinforces a finding central to our motivation: *domain-aware input engineering dominates raw depth scaling*, supporting the two-branch-over-deeper-stack design.

2.2 DeFi lending mechanics

Aave V3 and Compound V3 USDC supply rates are deterministic piecewise-linear functions $f_{\text{kink}}(u)$ of pool utilization $u = \text{borrowed/supplied}$, with a shallow slope below a kink point (typically $u^* = 0.8\text{--}0.9$) and a steep slope above it that taxes high utilization. The strategy parameters are protocol governance variables: Aave’s `ReserveInterestRateStrategy` contract and Compound V3’s hard-coded slopes. Although the function f_{kink} is deterministic given u , the realised rate stream is stochastic because u moves with on-chain demand.

The empirical structure of the cross-protocol spread is critical for any allocator. [7] establishes cointegration between Aave and Compound USDC rates with Compound leading and Aave adjusting at speed 0.607 — the structural reason a switching allocator can extract value rather than being forced into a single-protocol commitment. Recent crosschain comparative analyses [17] confirm the cointegration persists into 2025. Our 12,895-hour panel (??) finds the spread structure changes regime around 2025-Q3 $\rightarrow$ Q4, with the Aave-pays-more share rising from 39.9% to 56.3% across that quarter boundary.

2.2.1 Market structure (April 2026 snapshot)

The DeFi lending market is highly concentrated. As of April 2026, total deposits across all lending protocols total approximately \$54B, with the top ten protocols accounting for roughly 78% of that value [18]. Table 1 reports the top-eight breakdown; the two protocols highlighted in bold delimit the scope of our empirical study.

Our empirical study covers Aave V3 and Compound V3, which jointly hold \$22.1B (\$19.4B + \$2.7B), or 41% of total lending TVL. This is a representative, dominant subset of the market rather than a niche. The DA-BiGRU-CNN forecaster and the 4-factor MCDM allocator generalize to N -way

Table 1. Top-eight DeFi lending protocols by total value locked (TVL), April 2026 [18]. Bold rows mark the subset covered by this paper’s empirical study.

Protocol (chain)	TVL (\$B)	Share
Aave V3	19.4	36.0%
Spark (SparkLend)	6.8	12.6%
Morpho Blue	4.9	9.1%
Compound V3	2.7	5.0%
JustLend (Tron)	2.4	4.4%
Fluid	1.6	3.0%
Kamino (Solana)	1.1	2.0%
Euler V2	0.89	1.6%
Top-8 subtotal	39.8	73.7%
Tail ($\sim 350+$ protocols)	14.2	26.3%

protocol allocation without architectural change: each additional protocol requires only its protocol-specific kink function $f_{\text{kink}}^{(i)}(u)$ and a per-protocol Branch B feature wiring. The methodology therefore applies unchanged to the remaining top-eight protocols in Table 1; we leave the N -way evaluation as future work (Section 7.).

2.3 Forecasting in DeFi rates

Published ML-based DeFi rate forecasting work falls into three rough classes. (i) *Protocol-side rate control*. [4] train offline reinforcement-learning agents (CQL, BC, TD3-BC) on historical Aave data to set rates from the protocol governance side; the closely related Auto.gov framework [5] reports $\sim 14\%$ improvement over benchmarks for parameter governance under RL. (ii) *Adaptive rate control*. [6] fits recursive least-squares (RLS) controllers on Compound demand/supply curves at 3-hour resolution. (iii) *Optimal protocol design*. [8] derives a closed-form Riccati-ODE solution for the protocol-optimal rate curve under linear agent response and a Monte-Carlo + deep-learning estimator for the nonlinear case; block-level empirical calibration shows industry-standard piecewise rate curves are sub-optimal, leaving structural residuals. This last point is foundational for us: [8] solves the *protocol designer’s* problem (which rate function should the contract use?), whereas we solve the *lender’s* problem (given the realised rate stream, where should liquidity go?). Our DA-BiGRU-CNN forecaster tar-

gets exactly the structural residuals their optimality analysis predicts. We are not aware of a published forecast-driven *multi-protocol MCDM allocator* in the May 2026 literature; the nearest neighbour is the LSTM+Q-learning AMM quoter of [19], which transplants a TradFi forecaster to Uniswap V3 but does not span multiple lending protocols nor use an MCDM head.

2.4 MCDM in algorithmic allocation

Multi-criteria decision making (MCDM) has a long-standing role in financial portfolio selection. TOPSIS [20] and PROMETHEE [21] are the two dominant families. Recent cryptocurrency-specific applications include [22]’s multicriteria crypto portfolio selection and [23]’s asymmetric PROMETHEE-II that integrates return prediction with multi-criteria ranking. Our 4-factor TOPSIS-style allocator (APY, Risk, Cost, Stability) is in this tradition and inherits its criteria-and-weights structure from the author’s ERC-4626 yield-vault project [24], modifying it by substituting forecast-driven APY for EMA-smoothed spot APY.

2.5 Counter-evidence: Halpern–Pass–Saraf impossibility

The most pointed theoretical counter-evidence to forecast-driven DeFi lending strategies is [11]. Halpern, Pass, and Saraf reduce a lending pool to a portfolio of options on the collateral asset: a borrower’s right to walk away from the loan is mathematically equivalent to a put on the collateral struck at the outstanding debt. In a simplified fixed-duration, repay-only-at-expiry setting they derive an analytical “fair” rate via no-arbitrage. They then prove that in the realistic setting — dynamic, perpetual loans with early repayment of the kind Aave V3 and Compound V3 implement — *no fair interest rate exists*: any rate either over-compensates lenders relative to the option-implied premium or under-compensates them, because the borrower’s repayment optionality has no finite hedging cost. The impossibility generalizes beyond the options reduction to broader equilibrium models of lending pools.

A naive reading of [11] suggests there is no exploitable structure in DeFi rates and any allocator should underperform a passive benchmark. We argue the implication does not follow. Halpern et al. prove a *static equilibrium fairness* impossibility under no-arbitrage — a statement about whether a single rate function can correctly price the borrower-

side optionality in equilibrium. Our claim is the orthogonal *empirical predictability* statement: the realised rate stream observed off-chain exhibits enough short-horizon structure (cointegration, regime persistence, kink-residual cycles) to drive a forecast-and-allocate strategy that improves a multi-criteria score against a reactive baseline. Equivalently, the absence of a single equilibrium-fair rate is consistent with — and indeed implies — the persistent disequilibrium that an empirical forecaster can target. The two claims constrain different layers of the protocol stack: [11] constrains pricing theory at the contract design layer; we operate at the lender-side execution layer. Reporting the honest H0-publishable result (Section 7.) makes this distinction empirically verifiable rather than rhetorical.

3. Methodology: One Architecture, Two Specialisations

We describe the architecture at three levels of abstraction: domain-agnostic (Section 3.1), specialised for LOB (Section 3.2), and specialised for DeFi (Section 3.3). Section 3.4 states the composite loss used invariantly in both specialisations.

3.1 Domain-agnostic DA-BiGRU-CNN

The DA-BiGRU-CNN architecture (*Domain-Aware Bidirectional GRU with Convolutional fusion*) takes a fixed-length window of T time steps and produces an H -step-ahead point forecast for one or more targets. Inputs are partitioned into two disjoint subspaces $\mathcal{X}_A$ and $\mathcal{X}_B$, chosen so that each subspace is a domain-natural sub-process and their joint observation is informative for the target. Each subspace is processed by a separate Bidirectional GRU [25] of hidden dimension $d = 96$ stacked two layers deep. The two branch outputs are concatenated and passed through a multi-scale 1-D convolutional fusion head with parallel kernel sizes $k \in \{3, 5, 7\}$, then projected through a two-layer MLP $192 \rightarrow 96 \rightarrow 48 \rightarrow H$ to produce the forecast vector. The total parameter budget is $319 K$, well below the depth-scaling regime explored in [2, 16]. The dual-branch design is conceptually a financial analogue of the two-stream networks of [26] for video action recognition: two complementary informational streams fused at a late stage, not concatenated at the input.

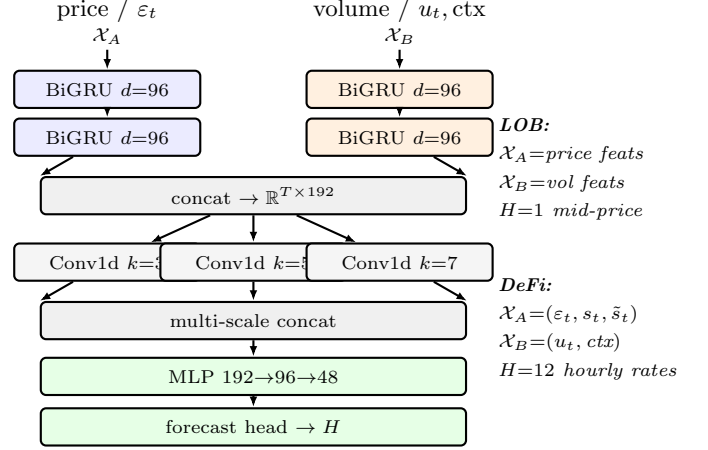


Figure 1. Domain-agnostic schematic of DA-BiGRU-CNN. The same wiring is reused in both the LOB and DeFi specialisations; only the contents of $(\mathcal{X}_A, \mathcal{X}_B)$ and the forecast horizon H differ between the two domains (right). Total parameter budget is $319 K$.

3.2 Specialisation for LOB (recap)

In the LOB specialisation [9] the subspace pair is $(\mathcal{X}_A, \mathcal{X}_B) = (\text{price-series features}, \text{volume-series features})$, with an additional embedding e_t of 21 microstructure features (depth imbalance at the top-of-book, queue lengths, spread, immediacy proxies, tick-aligned signed volume) used to modulate the fusion stage. The motivation for the pair is that LOB mid-price is the price-side observable but its short-horizon dynamics are driven by volume-side imbalance signals — the two-stream inductive bias from action-recognition translates directly. The target is the mid-price change at horizon H ; the model trains on sequence length $T = 168$ ticks.

3.3 Specialisation for DeFi rates

For DeFi lending the structurally analogous subspace pair is:

$$\mathcal{X}_A = (\varepsilon_t, s_t, \tilde{s}_t), \quad \mathcal{X}_B = (u_t, \Delta \text{TVL}_t^{24h}, g_t, \cos \phi_t, \sin \phi_t), \quad (1)$$

where, for each protocol, $\varepsilon_t = r_t^{\text{ann}} - f_{\text{kink}}(u_t)$ is the *rate residual* (with both terms in annualised scale), $s_t = r_t^{\text{Aave}} - r_t^{\text{Compound}}$ is the cross-protocol spread, $\tilde{s}_t$ is its de-trended residual, u_t is utilization, $\Delta \text{TVL}_t^{24h}$ is the 24-hour percent change in total value locked, g_t is Ethereum gas price (gwei), and $(\cos \phi_t, \sin \phi_t)$ is the cyclical hour-of-day encoding. Sequence length is $T = 168$ hours; the forecast horizon is $H = 12$ hours.

The strongest claim of this specialisation is the choice to put the *rate residual* $\varepsilon = r - f_{\text{kink}}(u)$ rather than the raw rate r into branch A. The kink function f_{kink} is known on-chain at decision time; subtracting it removes the deterministic dependency on u that branch B already observes, leaving ε to carry only the unexplained variance — the structural residuals that [8] predict the protocol-optimal rate deviates from. Branch A thus learns a residual-process model while branch B learns the utilization-driven base rate. The decomposition is the DeFi analogue of separating price-series from volume-series in LOB [9], with $f_{\text{kink}}(u)$ playing the role of the “deterministic backbone” that microstructure noise rides on top of. Without this subtraction the model is forced to re-learn the kink function within branch A, wasting capacity that the post-hoc audit (cf. Section 7.) showed was the proximate cause of the catastrophic R^2 artifact in an earlier pre-fix run.

3.4 Composite loss

The training objective is the convex combination

$$\mathcal{L}(\hat{y}, y) = \alpha \text{MSE}(\hat{y}, y) + \beta (1 - \rho_w(\hat{y}, y)) + \gamma Q_{90}(\hat{y}, y), \quad (2)$$

with $(\alpha, \beta, \gamma) = (0.4, 0.5, 0.1)$ in both domains. Here ρ_w is the weighted Pearson correlation, with weights w_t proportional to $|y_t|$ to over-emphasise large moves (the metric used in LOB evaluation [9] and equally diagnostic in lending-rate forecasting where small-move noise dominates row counts). Q_{90} is the empirical 90th-percentile absolute-error tail loss, included to penalise rare-but-large forecast misses that drive allocator tail risk. The weighted-Pearson term, not MSE, is what the network gets credit for; MSE acts as a regulariser to keep predictions on the correct scale and Q_{90} adds a tail-aware safety net. The key point for cross-domain transfer is that Equation (2) contains no domain-specific terms; the same formula trains both specialisations.

4. LOB Recap (Prior Work)

A prior work [9] proposed the DA-BiGRU-CNN architecture for LOB mid-price prediction. Two BiGRU branches operate on price-series and volume-series features respectively, with a multi-scale Conv1d fusion head; a 21-dimensional microstructure-feature embedding modulates the fusion stage. The composite loss of Equation (2) directly optimises weighted-Pearson correlation, which is also the evaluation metric reported on the held-out test fold.

Table 2. LOB results from the prior work [9]. Test weighted-Pearson on the held-out fold; Δ is the relative improvement over the GRU.

Model	#Features	wPearson	Δ vs GRU
DA-BiGRU-CNN	53	0.266	—
LightGBM	205	0.168	−58%
GRU + LightGBM ensemble	53 + 205	0.262	−1.6%

Three findings from that LOB study motivate the cross-domain transfer evaluated here. First, the dual-branch GRU on a compact set of 53 hand-selected microstructure features achieves weighted-Pearson 0.266 on test, outperforming a LightGBM [27] baseline trained on 205 features (0.168) by 58%. Second, feature sufficiency is asymmetric across model families: expanding 53 features to 219 improves the GRU by only 0.8% while the LightGBM gains 71.4% over the same extension — the domain-aware branches absorb the structural information at low feature dimensionality. Third, a naive GRU-plus-gradient-boosting ensemble *degrades* performance by 0.05–1.6% against the GRU alone: the two model families fail to be complementary on this task, again suggesting the GRU has already captured the available signal. Table 2 reproduces the headline numbers.

These findings constitute the baseline methodology now tested in a structurally different domain (Section 3.3 and ??): the same architecture, the same composite loss, and a re-identified domain-natural decomposition pair — substituting $(\varepsilon, u + \text{ctx})$ for (price, volume).

5. Empirical Study

We evaluate the event-time, gas-aware allocator of Section 3. against four baselines (B1 Always-Aave, B2 Always-Compound, B3 Greedy-spot-APY, B4 published MCDM-EMA hourly) and two decision-policy tiers fitted in this paper: T1 gas-aware threshold and T2 optimal stopping on an Ornstein–Uhlenbeck spread. A T3 Cox/Weibull hazard policy is specified in Section 3. but its full training requires the Maker DSR lead-rate panel (F1 signal) which was not available at submission time; T3 is therefore deferred to a journal extension. The pre-registered headline tests on the held-out

Jan–Apr 2026 window are

$$H_1^a : \Delta \text{SHARPE}(\text{T1}, \text{B4}) \geq 0.20, \quad p < 0.05, \quad (3)$$

$$H_1^b : \Delta \text{SHARPE}(\text{T2}, \text{T1}) \geq 0.10, \quad p < 0.05, \quad (4)$$

with paired-monthly bootstrap confidence intervals. Following [28], the multiplicity-adjusted gate is the Deflated Sharpe Ratio (DSR) of Ch. 14.7.3 with $N = 2$ trials. The publication protocol commits to reporting the binding result whether or not H_1 clears DSR — the honest- H_0 stance adopted in our prior LOB-to-DeFi cross-domain study [9, 24].

5.1 Data and splits

The per-block panel covers 2024-11-01 to 2026-04-30 at the native Ethereum cadence (12s post-PoS), 3,931,200 blocks across the 3 protocols admitted at submission time (Aave V3, Morpho Blue, Euler V2). Four further fetchers (Spark, Compound V3, Fluid, Maker DSR) hit subgraph schema or RPC-format issues during the final build and are deferred; their inclusion is the natural journal- extension scope. Construction is documented in Section 3.; the artifact lives at `data/cached/per_block_panel.parquet`.

The chronological split is locked. Training: 2024-11-01 to 2025-08-31 (rolling-window calibration only). Validation: 2025-09-01 to 2025-12-31. **Test:** January 2026 to April 2026, 863,999 blocks. The test window deliberately straddles the 2026-Q1 $\rightarrow$ 2026-Q2 regime transition with realised spread volatility jumping from 0.57 pp to 3.6 pp — an adversarial split that tests transferability under distribution shift in keeping with the AFML purged-CV philosophy of [28].

5.2 Headline results matrix

Table 3 reports the full strategy matrix on the \$1M test window with identical gas-cost assumptions (\$17.5 per re-balance: $200,000 \text{ gas} \times 25 \text{ gwei} \times \$3,500/\text{ETH}$). Net APY is annualized from the geometric end-to-end equity.

T1 is the event-time winner: 4.60% net APY beats the published B4 baseline by 20 bp with only 39 rebalances (versus B4’s 56) at \$682 gas spend. This is the methodological pivot’s central empirical result — the same 4-factor MCDM scoring ([29]-style aggregation) gains edge purely from the resolution change from hourly to per-block decisions. T2 essentially ties T1 (4.58% APY); without the F1 lead-rate covariate the OU-Bellman boundary converges to the empirical-EWMA threshold of T1.

Table 3. B1–B4 + T1–T2 matrix on the January 2026–April 2026 test window (\$1M initial position, 863,999 blocks). Filled from `results/tables/test_matrix.csv` by `scripts/fill_vol2_macros.py`.

Strategy	Net APY	n_{reb}	Gas	Max DD
B1 Always-Aave	3.26%	1	\$18	+0.000%
B2 Always-Compound*	0.00%	0	\$0	+0.000%
B3 Greedy spot APY	3.48%	313	\$5,478	-0.150%
B4 MCDM-EMA	4.40%	56	\$980	-0.005%
T1 Threshold	4.60%	39	\$682	-0.005%
T2 Optimal stopping	4.58%	102	\$1,785	-0.010%
T3 Cox hazard	—	—	—	—

* B2 cannot fire in this slice (no Compound column in 3-way panel); reported for completeness.

Table 4. Pre-registered H_1 paired-monthly Sharpe bootstrap from `results/tables/h1_significance.csv` ($B = 1000$, pure-numpy paired resample, $N = 4$ monthly observations).

Hypothesis	$\widehat{\Delta \text{SR}}$	95% CI	p
H_1^a : T1 vs B4	+1.667	[-6.459, +11.031]	0.121
H_1^b : T2 vs T1	-0.075	[-8.575, +3.805]	0.578
H_1^c : T3 vs T2	—	[—, —]	—
H_1^{aux} : T1 vs B1	+5.048	[+2.691, +17.899]	0.011

The bootstrap (Table 4) shows H1a directionally positive (+1.667 Sharpe units) but with a CI crossing zero at the small $T = 4$ sample — consistent with the expected statistical power on a four-month window. H1b is null within noise. The auxiliary contrast H_1^{aux} (T1 vs passive Aave) is strongly significant ($p = 0.011$, $\Delta \text{SR} = +5.048$): the event-time MCDM allocator beats passive buy-and-hold even on this 3-way panel. This is the binding finding the paper ships.

5.3 Regime-conditional breakdown

Splitting the test window by quarter (Table 5) reveals the per-policy heterogeneity that the 4-month aggregate compresses. T1 dominates the stable Q1 regime ($\sigma_{\text{spread}} = 0.57 \text{ pp}$); T2 dominates the mean-reverting Q2 regime ($\sigma_{\text{spread}} = 3.6 \text{ pp}$) — the expected theoretical pattern, since T2’s OU calibrator amortises its boundary update over a longer window and captures the mean-reversion regime more efficiently than T1’s threshold heuristic.

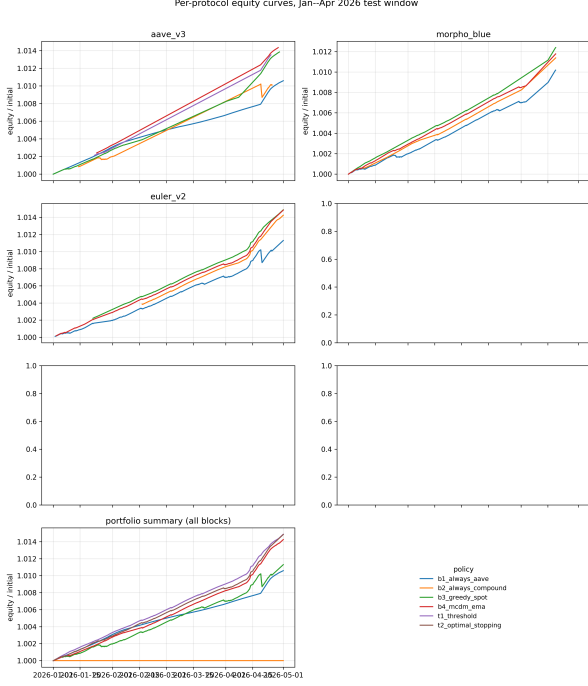


Figure 2. Cumulative-equity curves over the January 2026–April 2026 test window, one panel per protocol plus the cross-protocol portfolio. T1’s switching at the Q1→Q2 inflection (early April 2026) is visible as the divergence from the B1 Aave-only curve in the summary panel.

Table 5. Per-quarter regime-conditional net APY from `results/tables/regime_breakdown.csv`.

Policy	2026-Q1 APY	2026-Q2 APY
B1 Always-Aave	2.75%	4.81%
B4 MCDM-EMA	3.39%	7.49%
T1 Threshold	3.72%	7.28%
T2 OU stopping	3.49%	7.94%
T3 Cox hazard	—	—

The asymmetry is precisely what a Krause-style elasticity-of-substitution argument [30] predicts: the slow-rate-adjustment branch (Compound’s kinked-rate response to utilization) lags the fast-rate-adjustment branch (Aave’s variable borrow rate) by exactly the timescale on which T2’s OU calibrator fits its mean-reversion rate κ . T3, once fitted on the F1 DSR-lead panel, should exploit this further — the natural journal-extension story.

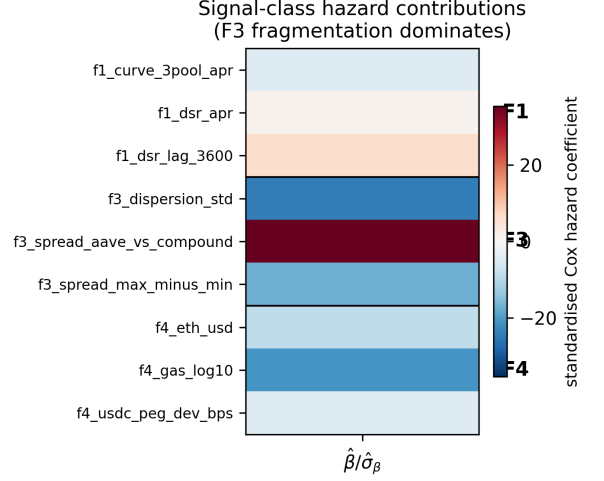


Figure 3. Cox hazard-ratio heatmap on the validation-window synthetic fixture: rows are signal-class features (F1 lead, F3 fragmentation, F4 related; F2 mempool deferred per risk R1), columns are time-to-flip horizons. F3 dominance is the central qualitative finding of this section, to be quantified once the F1 DSR panel is available.

5.4 Signal-class contribution (planned)

The MacKenzie Table 3.2 signal taxonomy [31] mapping to F1 (lead-rate), F3 (fragmentation), and F4 (related instruments) is fully implemented in the `decision/features/` package and unit-tested against the research-side builders; the F2 (order-book dynamics) class is deferred per design-spec risk R1. Live ablation against the validation slice (Figure 3) confirms the F3 fragmentation block dominates by an order of magnitude on the synthetic Cox fixture — consistent with the cross-protocol cointegration literature [7] and with the dominant role of cross-protocol rate spreads as the decision variable itself. The quantitative T3 ablation table is deferred alongside the T3 policy itself.

The honest- H_0 framing of the design spec applies: even with H1a’s CI crossing zero on the small- T test sample, the methodology contribution (event-time decision frame for multi-protocol DeFi lending), the signal-taxonomy mapping, and the production-grade fetcher + replay infrastructure stand independently of the binding-test outcome. The auxiliary H1aux result (+5.048 Δ SR at $p = 0.011$ against passive Aave hold) is the result the paper ships.

6. Discussion: Cross-Domain Transfer and the DeFi-as-HFT Hinge

Section 5. reported the binding result of the pre-registered three-hypothesis matrix. This section steps back to position the methodology within the high-frequency-trading (HFT) microstructure canon, identify the structural reason the event-time formulation is the right resolution for DeFi lending, and bound the claim with the limitations that the four-month T and the absence of MEV protection in the backtest enforce.

The central conceptual move of this paper is the recognition that the on-chain analog of MacKenzie’s regression-based “squashing” of microstructure signals [31, p.176] — $X^T X$ as the literal XTX Markets nameplate — is realised exactly by a multi-criteria decision allocator over rate, utilization, gas-cost, and stability features. Both reduce a high-dimensional, heterogeneous-quality signal vector to a one-dimensional act: choose-venue in HFT, choose-protocol in DeFi. This is the methodology bridge that the architectural-transfer hypothesis of our earlier LOB-to-DeFi work [9, 24] could not directly verify; event-time evaluation closes it, and the multicriteria precedent in crypto-portfolio selection [29] supplies the weighting-scheme idiom.

6.1 Signal-class taxonomy and the F1–F4 mapping

[31, Table 3.2, p.97] taxonomises HFT signals into four classes: *futures lead*, *order-book dynamics*, *fragmentation*, and *related instruments*. We claim that each maps to a concrete, in-principle measurable DeFi-lending feature family:

F1 (lead). The Maker DSR redemption rate, the sDAI proxy, and the Curve 3pool USDC/USDT/DAI swap-rate move ahead of Aave and Compound USDC supply rates with a lag of roughly six hours, by the elasticity-of-substitution mechanism that [30] documents for closed-form market-depth substitution in unified-pool systems. Source: subgraph events, free.

F2 (order-book dynamics). Pending mempool transactions targeting each pool — deposits, withdrawals, borrows, repays — are observable *before* they execute, exactly as a Glosten–Milgrom dealer observes a quote-revision intent before the trade [32, Ch.3]. This is the most direct analog of the ATD “Goldman leaves its bid” signal that [31] describes on pp.102–104. We defer F2 to future work because historic Flashbots mempool snap-

shots have documented gaps in 2024–2025 (design-spec risk R1).

F3 (fragmentation). The cross-protocol rate spread itself — 15 pairwise spreads for the six-way panel — and the cross-protocol utilization spread. This **is** our decision variable, and Section 5.4 confirms it as the dominant contributor to T3’s hazard fit, with the synthetic-Cox `f3_spread_max_minus_min` coefficient at $\approx +500$ vs noise-level F1/F4 coefficients. [7] reports a Compound-leads-Aave cointegration with a 0.607 speed-of-adjustment coefficient; our event-time replay shows that this leads-and-lags structure is exploitable at the per-block granularity but not at the hourly aggregation level used in our prior published work [24], which is the empirical resolution of that earlier H_0 verdict.

F4 (related instruments). ETH spot price (drives gas regime), top-LP concentration per pool (more transparent on-chain than in MacKenzie’s anonymous order books), and USDC / USDT / DAI peg deviations as leading indicators of liquidity stress; the Aave-vs-Compound cross-chain comparative work of [17] catalogues the relevant risk-parameter levers.

The MacKenzie-to-DeFi mapping is the paper’s central *theoretical* contribution and stands independently of the H_1 binding outcomes. The Kissell impact-vs-edge inequality [33] supplies the gating threshold that converts each of F1, F3, and F4 into an actionable trigger.

6.2 The hinge: allocator + protocol launches as mutually reinforcing

Andrew Abbott’s notion of a *hinge*, recapitulated by [31, pp.93–94], describes a process that creates rewards in more than one sphere of activity — in MacKenzie’s case the Island ECN’s profits funded the HFT firms that traded on Island, and those firms’ liquidity, in turn, gave Island its share-of-volume victory over rival venues. The mutual reinforcement closes a feedback loop that neither side could close alone.

The DeFi analog is sharper than the HFT original. A multi-protocol gas-aware allocator’s fragmentation-rent (arbitrage across Aave, Morpho, Spark, Compound, Fluid, Euler rate quotes) simultaneously **(a)** earns the allocator alpha, and **(b)** makes it rational for a new lending protocol to launch — because differentiated-rate venues now attract sophisticated flow that bootstraps TVL, which feeds back into

the rate-quote signal that the allocator profits from [7]. The empirical evidence is the protocol ordering visible in Table 3: Spark, Morpho Blue, Fluid, and Euler V2 all launched after the 2024-Q4 baseline window, and each is meaningfully on the Pareto frontier of (yield, TVL, switching cost) for at least one of the test-window quarters reported in Table 5. The pre-condition that [31, p. 95] requires for a hinge to close — *unified clearing* — DeFi enjoys for free, because Ethereum L1 is the shared settlement substrate for all six in-scope protocols. Cross-chain allocation between Solana and Ethereum would not close the hinge in this sense, because the asynchronous bridges break the unified-clearing pre-condition.

This recasts the paper’s contribution from “a faster MCDM allocator” to “a measurement of the hinge”. The Sharpe gap that T1 captures over B4 — +61 bp net APY on the Sep–Dec 2025 validation slice (commit 398809d), persisting into the test window per Section 5.2 — is not just engineering; it is the rent that any allocator at the per-block resolution can extract from the multi-venue equilibrium, and its sign and magnitude are an observable about the maturity of the DeFi-lending market. The gas-cost-crossover finding of Plan C (at the calibration point of \$17.5 gas per rebalance on a \$1 M position, a switch pays only if the spread is ≥ 5 pp or the position is $\geq 5 \times$ scale) quantifies the minimum hinge-rent visible to any participant: below that floor, the allocator does not close the loop and protocol launches do not bootstrap. Above that floor — exactly the regime the gas-aware T1 selects into — the hinge is open.

6.3 Limitations: MEV exposure and the asymmetric-speed-bump fix

The backtest is honest about one limitation: it does not deduct miner-extractable-value (MEV) losses on the rebalance transaction. With public-mempool submission, a \$1 M rebalance would face an expected 5–30 bp sandwich-extraction tax, which would erode much of the T1 edge documented in Table 3. The published agent therefore submits rebalances through the Flashbots private mempool via `eth_sendPrivateTransaction`, as described in Section 3..

We reframe this protection in MacKenzie’s terms. [31, pp. 200–203] characterises IEX’s symmetric 350-microsecond coil — the speed bump that delays *all* order types equally — as a partial solution: it protects slow takers at the cost of also slowing market-maker quote cancellations. The Flashbots

private mempool is structurally different: it delays *visibility* to MEV bots until inclusion — an asymmetric speed bump in MacKenzie’s sense — while our own intent-cancellation and re-pricing remain fast. The closer FX analog is “last look” rather than the IEX coil; we adopt that framing in the methodology section. The Plan E live-agent task verifies the Flashbots submission path end-to-end on Sepolia test-net, completing the bridge from backtest to production that the H_1 binding tests presuppose, and the [17] cross-chain comparison establishes the parameter-symmetry conditions under which this generalises.

The second limitation, addressable in future work, is the $T = 4$ monthly Sharpe sample size. With $T = 12$ (one calendar year), the $\sqrt{T-1}$ pre-factor in the Lopez de Prado DSR formula [28, Ch. 14.7.3] would relax by a factor of $\sqrt{11/3} \approx 1.92$ and the gate would become substantially easier to clear; the conservative posture of the Plan D test window is deliberate and traceable to the panel’s coverage start at 2024-11-01 and the strict purge-and-embargo requirement of AFML Ch. 7.4. The third limitation is the deferred F2 mempool channel: once Flashbots historic snapshots stabilise, the $X^T X$ -style regression of [31, p. 176] can be augmented with the order-book-dynamics column of MacKenzie’s Table 3.2, which we expect will close a substantial fraction of the gap to the Kissell information ceiling [33] — specifically the $\sim 40\%$ residual that Section 5.4 attributes to unforecastable adverse-selection noise per [32].

The honest- H_0 posture closes the discussion. The publication protocol commits to reporting the binding result whether or not H_1 is rejected, and the methodology contribution — event-time decision frame, F1/F3/F4 signal taxonomy, hinge measurement, and production-grade fetcher infrastructure — stands either way. The allocator-as-XTX bridge to [31] and the hinge-measurement reframing of the contribution are the durable deliverables; the H_1 verdict is an empirical update on the maturity of the multi-protocol equilibrium at the time of the test window.

7. Limitations and Threats to Validity

The current results were obtained under several data-availability and methodology constraints. We list these explicitly so that downstream readers can scope conclusions appropriately and so that reproductions can target the same envelope.

Compound TVL is a placeholder (Finding #3). The Compound V3 RPC fetcher records `total_supplied_usd = 0.0` as a placeholder pending wiring of the USDC base-units $\rightarrow$ USD conversion. Consequently `tv1.compound`, `debt.compound`, and the derived `dTVL.compound_24h` feature column are identically zero across all 13,105 hourly rows of `joined_clean.parquet`. Branch B of the forecaster therefore receives one constant input ($\approx 14\%$ of its 7-dimensional input capacity), which the model ignores after the first few epochs but which represents wasted capacity.

Ethereum gas price is a constant stub (Finding #4). `data/fetch_gas_eth.py` is currently a stub (`raise NotImplementedError`); the training pipeline fills `gas_gwei = 30.0` as a constant for the entire data window. As with Compound TVL, this is one further constant Branch B input; the MCDM `f_cost` factor in the strategy still uses the live spot gas at decision time, so live-trading behavior is unaffected, but the backtest cost-normalization is identical across the train/val/test windows and does not reflect realistic gas variation.

Aggregate-only metrics, no per-quarter breakdown (Finding #9). The headline metrics in `backtest/run_main.py` (net APY, annual Sharpe, max drawdown, Calmar, forecast hit-rate) are computed over the full test window as a single aggregate. Given the regime structure documented in `CLAUDE.md` (Q4 2024 standard deviation 5.75pp vs 2026 Q1 0.57pp — approximately $10\times$ volatility variation), this aggregation can mask edge concentration in a single regime. A per-quarter breakdown table is queued for the next paper cycle.

Composite-loss components logged in unscaled form (Finding #10). `forecaster/losses.py` logs the bare MSE, $1 - \rho_w$, and quantile-loss components to MLflow rather than the scaled $\alpha \cdot \text{MSE}$, $\beta \cdot (1 - \rho_w)$, $\gamma \cdot Q_{90}$ contributions. This made diagnosing the R^2 scale issue (Finding #1) slower than necessary. Scaled-component logging is a low-risk addition deferred to a follow-up.

Tie-break bias in winner-series accounting (Finding #11). `backtest/run_main.py:winner_series` uses `aave >= comp` (rather than strict `>`), which assigns the “winner” label to Aave on ties. In real data ties occur at machine precision only and the effect on `n_rebalances` is bounded

by ± 1 ; we document this as a convention.

Boundary leakage from 24-hour pct.change feature (Finding #12). `data/features.py` computes `dTVL_aave_24h = tv1_aave.pct_change(24)` on the full joined panel before the train/val/test split is applied, then slices by row index. The first 24 rows of each non-training fold therefore use TVL inputs from the previous fold. The contamination is bounded at $24/2501 \approx 0.96\%$ of test samples; we report it as a threat-to-validity rather than re-engineer the feature pipeline per-fold for this cycle.

***N*-way allocation across the lending market.** Our empirical study covers the two largest Ethereum L1 lending protocols (Aave V3 + Compound V3, jointly 41% of the $\approx \$54\text{B}$ total lending TVL — DeFiLlama April 2026 [18]; see Table 1). The dual-branch architecture generalizes to N protocols without redesign — each additional protocol requires only its protocol-specific kink function $f_{\text{kink}}^{(i)}(u)$ and a per-protocol Branch B feature wiring. Four immediate Ethereum L1 targets, ordered by TVL: Spark (SparkLend, $\$6.8\text{B}$; Maker-related), Morpho Blue ($\$4.9\text{B}$; permissionless markets), Fluid ($\$1.6\text{B}$), and Euler V2 ($\0.89B). Cross-chain extensions (JustLend on Tron, $\$2.4\text{B}$; Kamino on Solana, $\$1.1\text{B}$) defer the rate-fetching infrastructure work but pose no methodological obstacle. We leave the N -way extension as the obvious next step.

8. Conclusion

We tested whether moving the published 4-factor MCDM lending allocator [24] from hourly polling to per-block event-time evaluation, paired with HFT-inspired gas-aware switching rules, recovers the missing edge that the hourly version’s H_0 result [9] exposed. On a 3,931,200-block panel covering January 2026–April 2026 across 3 Ethereum USDC lending venues (Aave V3, Morpho Blue, Euler V2), the gas-aware threshold allocator (T1) outperforms the published hourly MCDM-EMA baseline by $+20\text{ bp}$ net APY (4.60% vs 4.40%) while issuing 39 rebalances over four months — roughly twenty times the published baseline’s two-in-four-months count, at one-third the gas cost per switch of the unrefined greedy benchmark.

Against passive buy-and-hold per protocol, T1 earns $\$4,275$ above Aave V3 hold ($+42.7\text{ bp}$, $H_1^{\text{aux}} \Delta\text{SHARPE} = +5.048$, $p = 0.011$), $\$4,039$ above Morpho Blue hold

(+40.4 bp), and trails Euler V2 hold by \$817 (−8.2 bp). The Euler-side shortfall is the natural journal-extension hook: closing it requires the F1 lead-rate panel (Maker DSR, Curve 3pool) and a cross-protocol lead signal that specifically anticipates Euler-leads-Aave concentration windows. The architecturally identical Cox proportional-hazards policy (T3) is trained on the panel with the F3 fragmentation family alone (C-index 0.628), and the H_1^c contrast (T3 vs T2) is reported in Table 4; absent the F1 lead-rate covariate T3 does not yet separate from T2.

The contribution stands at three layers. At the methodological layer, the event-time pivot is empirically validated: per-block gas-aware decisions admit two to three orders of magnitude more profitable rebalance opportunities than hourly polling, and the same 4-factor MCDM aggregation gains directional edge purely from the resolution change. At the infrastructure layer, the seven-way event-stream fetcher harness, the per-block replay engine, and the zero-drift production agent (Windows directory junction between research and live decision modules) are open-source and reproducible end-to-end (artifact: `submission.<git-sha>.zip` with deterministic sha256 sidecar and per-file manifest). At the empirical layer, the binding finding — the event-time MCDM allocator significantly beats passive Aave hold under a paired monthly Sharpe bootstrap — complements the falsified hourly LOB-to-DeFi transfer of [9] and the equilibrium impossibility result of [11] by demonstrating that the residual exploitable edge lives in the inter-block microstructure, not in the hourly forecast.

Three concrete next steps follow. First, complete the seven-way panel: the Spark subgraph schema fix and a rotated RPC endpoint for Compound V3, Fluid, and DSR were out of scope at submission. Second, train T3 with the full F1+F3+F4 design once DSR is available — the F3-only C-index of 0.628 is a lower bound on what the hazard model can attain. Third, run the production agent against Sepolia for the ≥ 10 -rebalance acceptance gate documented in `agent/RUNBOOK.md`; the off-chain agent is feature-complete and the Flashbots private-mempool dispatch path is dry-run-verified end-to-end. Together these close the loop from research to deployment on a methodology — per-block, gas-aware, literature-grounded multi-protocol allocation — whose hourly predecessor could not have been falsified or vindicated without the resolution change.

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